

Minimal block VI: the freezing limit

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Abstract

The right-hand side of the freezing inequality (unit 77) is normalised by $N^p / \log^{d_0+1} N$: the strip terms vanish because the mixed-block envelope loses one logarithm, the corner term converges to ωC_4 by the frozen coefficient theorem, and uniform continuity of ξ, η on the closed box makes ω arbitrarily small. Hence $N^p (Z(N) - Z_0(N)) / \log^{d_0+1} N \rightarrow 0$: the leading coefficient of a block of multiplicity $m = d_0 + 2 \geq 2$ depends on ξ, η only through their values on the minimal face $u = 0$. Zero sorry/axiom.

1 The things formalised

```
theorem corner_oscillation {m d' : } (b : ) (hb : 0 < b) ( ) (hc : Continuous ...)
  ( : ) (h : 0 < ) :
    > 0, u v, ( i, 0 u i u i b) → ( i, 0 v i v i b) → ( i, u i ) →
    | u v - 0 v|
theorem constStateIntegral_shift_tendsto_zero ... (hp : j, finExp k h j = p) ... (i) :
  Tendsto (fun N : => N ^ p / Real.log N ^ (d + 1)
    * constStateIntegral a b N k (shiftExp h i) k' h') atTop ( 0)
theorem blockStateIntegral_frozen_eq ... :
  blockStateIntegral b N k h k' h' (fun _ v => 0 v) (fun _ v => 0 v)
  = frozenStateIntegral b N (toNatFun k 1) (toNatFun h 0) d k' h' (fun v => 0 v)
  (fun v => 0 v)
theorem blockStateIntegral_sub_frozen_tendsto ( b p : ) (h : 0 < ) (hb : 0 < b)
  {d d' : } (k h : Fin (d + 2) → ) (hk : j, 0 < k j) (hp : j, finExp k h j = p)
  (k' h' : Fin d' → ) (hk' : i, 0 < k' i) (hq : i, p < ((h' i : ) + 1) / k' i)
  ( ) (hc : Continuous ...) (hc : Continuous ...) :
  Tendsto (fun N : => N ^ p / Real.log N ^ (d + 1)
    * (blockStateIntegral b N k h k' h'
      - blockStateIntegral b N k h k' h' (fun _ v => 0 v) (fun _ v => 0 v))) atTop
  ( 0)
```

2 Proof strategy

Strip terms: on the open box the shifted block is bounded by $K(Nv^{k'})^{-p(1+\log_+(Nv^{k'}B))^{d_0}}$, hence the normalised integrand by $K(2 + \log_+(B'B))^{d_0} v^{h'}(v^{k'})^{-p}/\log N$; `integral_mono_of_nonneg` needs only integrability of the dominating side, and `squeeze_zero'` against $C/\log N$ finishes. Freezing limit: given ε , choose $\omega = \varepsilon/(2C_2(|C_4| + 1))$, get δ from uniform continuity (`IsCompact.uniformContinuousOn_of_continuous`, `Metric.uniformContinuousOn_iff`, with $\text{dist}((u, v), (0, v)) = \max_i |u_i|$ via `Prod.dist_eq` and `dist_pi_le_iff`), then eventually the corner term is $< \varepsilon/2$ and the strip sum is $< \varepsilon/2$.

3 Roadblocks and resolutions

- `Tendsto.congr'` pointwise goals are beta redexes: `beta_reduce` before `rw`.
- Rewriting the frozen inner integral with `← integral_const_mul` is ambiguous (two matching products); state the integrand identity by `integral_congr_ae` and rewrite forward.
- `field_simp` closes the two scalar identities outright; a trailing `ring` errors with “no goals”.
- `tendsto_finset_sum` is deprecated for `tendsto_finsetSum`.

4 Outcome

Astra target 5 is essentially complete: the general multiplicity- m leading term reduces to the frozen one. Next: assemble the main theorem $Z \sim C N^{-p} \log^{d_0+1} N$ with $C = \frac{1}{(d_0+1)! \prod k_i} \int v^{h'} (v^{k'})^{-p} A_{p-1}(\xi(0, v)) \eta(0, v) dv$, positivity when $\eta(0, \cdot) > 0$, the $\log n$ form, and the Fubini bridge to the full box.